

In Battle Over A.I., Meta Decides to Give Away Its Crown Jewels

The tech giant has publicly released its latest A.I. technology so people can build their own chatbots. Rivals like Google say that approach can be dangerous.



By Cade Metz and Mike Isaac

Cade Metz, who covers artificial intelligence, and Mike Isaac, who covers Meta, reported from San Francisco.

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In February, Meta made an unusual move in the rapidly evolving world of artificial intelligence: It decided to give away its A.I. crown jewels.

The Silicon Valley giant, which owns Facebook, Instagram and WhatsApp, had created an A.I. technology, called LLaMA, that can power online chatbots. But instead of keeping the technology to itself, Meta released the system's underlying computer code into the wild. Academics, government researchers and others who gave their email address to Meta could download the code once the company had vetted the individual.

Essentially, Meta was giving its A.I. technology away as open-source software — computer code that can be freely copied, modified and reused — providing outsiders with everything they needed to quickly build chatbots of their own.

“The platform that will win will be the open one,” Yann LeCun, Meta's chief A.I. scientist, said in an interview.

As a race to lead A.I. heats up across Silicon Valley, Meta is standing out from its rivals by taking a different approach to the technology. Driven by its founder and chief executive, Mark Zuckerberg, Meta believes that the smartest thing to do is

share its underlying A.I. engines as a way to spread its influence and ultimately move faster toward the future.

Its actions contrast with those of Google and OpenAI, the two companies leading the new A.I. arms race. Worried that A.I. tools like chatbots will be used to spread disinformation, hate speech and other toxic content, those companies are becoming increasingly secretive about the methods and software that underpin their A.I. products.

Google, OpenAI and others have been critical of Meta, saying an unfettered open-source approach is dangerous. A.I.'s rapid rise in recent months has raised alarm bells about the technology's risks, including how it could upend the job market if it is not properly deployed. And within days of LLaMA's release, the system leaked onto 4chan, the online message board known for spreading false and misleading information.

"We want to think more carefully about giving away details or open sourcing code" of A.I. technology, said Zoubin Ghahramani, a Google vice president of research who helps oversee A.I. work. "Where can that lead to misuse?"

Some within Google have also wondered if open-sourcing A.I. technology may pose a competitive threat. In a memo this month, which was leaked on the online publication Semianalysis.com, a Google engineer warned colleagues that the rise of open-source software like LLaMA could cause Google and OpenAI to lose their lead in A.I.

But Meta said it saw no reason to keep its code to itself. The growing secrecy at Google and OpenAI is a "huge mistake," Dr. LeCun said, and a "really bad take on what is happening." He argues that consumers and governments will refuse to embrace A.I. unless it is outside the control of companies like Google and Meta.

"Do you want every A.I. system to be under the control of a couple of powerful American companies?" he asked.

“The platform that will win will be the open one,” said Yann LeCun, Meta’s chief A.I. scientist. Victor Llorente for The New York Times

OpenAI declined to comment.

Meta's open-source approach to A.I. is not novel. The history of technology is littered with battles between open source and proprietary, or closed, systems. Some hoard the most important tools that are used to build tomorrow's computing platforms, while others give those tools away. Most recently, Google open-sourced the Android mobile operating system to take on Apple's dominance in smartphones.

Many companies have openly shared their A.I. technologies in the past, at the insistence of researchers. But their tactics are changing because of the race around A.I. That shift began last year when OpenAI released ChatGPT. The chatbot's wild success wowed consumers and kicked up the competition in the A.I. field, with Google moving quickly to incorporate more A.I. into its products and Microsoft investing \$13 billion in OpenAI.

While Google, Microsoft and OpenAI have since received most of the attention in A.I., Meta has also invested in the technology for nearly a decade. The company has spent billions of dollars building the software and the hardware needed to realize chatbots and other "generative A.I.," which produce text, images and other media on their own.

In recent months, Meta has worked furiously behind the scenes to weave its years of A.I. research and development into new products. Mr. Zuckerberg is focused on making the company an A.I. leader, holding weekly meetings on the topic with his executive team and product leaders.

On Thursday, in a sign of its commitment to A.I., Meta said it had designed a new computer chip and improved a new supercomputer specifically for building A.I. technologies. It is also designing a new computer data center with an eye toward the creation of A.I.

"We've been building advanced infrastructure for A.I. for years now, and this work reflects long-term efforts that will enable even more advances and better use of this technology across everything we do," Mr. Zuckerberg said.

Meta's biggest A.I. move in recent months was releasing LLaMA, which is what is known as a large language model, or L.L.M. (LLaMA stands for "Large Language Model Meta AI.") L.L.M.s are systems that learn skills by analyzing vast amounts of

text, including books, Wikipedia articles and chat logs. ChatGPT and Google's Bard chatbot are also built atop such systems.

L.L.M.s pinpoint patterns in the text they analyze and learn to generate text of their own, including term papers, blog posts, poetry and computer code. They can even carry on complex conversations.

In February, Meta openly released LLaMA, allowing academics, government researchers and others who provided their email address to download the code and use it to build a chatbot of their own.

But the company went further than many other open-source A.I. projects. It allowed people to download a version of LLaMA after it had been trained on enormous amounts of digital text culled from the internet. Researchers call this "releasing the weights," referring to the particular mathematical values learned by the system as it analyzes data.

This was significant because analyzing all that data typically requires hundreds of specialized computer chips and tens of millions of dollars, resources most companies do not have. Those who have the weights can deploy the software quickly, easily and cheaply, spending a fraction of what it would otherwise cost to create such powerful software.

As a result, many in the tech industry believed Meta had set a dangerous precedent. And within days, someone released the LLaMA weights onto 4chan.

At Stanford University, researchers used Meta's new technology to build their own A.I. system, which was made available on the internet. A Stanford researcher named Moussa Doumbouya soon used it to generate problematic text, according to screenshots seen by The New York Times. In one instance, the system provided instructions for disposing of a dead body without being caught. It also generated racist material, including comments that supported the views of Adolf Hitler.

In a private chat among the researchers, which was seen by The Times, Mr. Doumbouya said distributing the technology to the public would be like "a grenade available to everyone in a grocery store." He did not respond to a request for comment.

Stanford promptly removed the A.I. system from the internet. The project was designed to provide researchers with technology that “captured the behaviors of cutting-edge A.I. models,” said Tatsunori Hashimoto, the Stanford professor who led the project. “We took the demo down as we became increasingly concerned about misuse potential beyond a research setting.”

Dr. LeCun argues that this kind of technology is not as dangerous as it might seem. He said small numbers of individuals could already generate and spread disinformation and hate speech. He added that toxic material could be tightly restricted by social networks such as Facebook.

“You can’t prevent people from creating nonsense or dangerous information or whatever,” he said. “But you can stop it from being disseminated.”

For Meta, more people using open-source software can also level the playing field as it competes with OpenAI, Microsoft and Google. If every software developer in the world builds programs using Meta’s tools, it could help entrench the company for the next wave of innovation, staving off potential irrelevance.

Dr. LeCun also pointed to recent history to explain why Meta was committed to open-sourcing A.I. technology. He said the evolution of the consumer internet was the result of open, communal standards that helped build the fastest, most widespread knowledge-sharing network the world had ever seen.

“Progress is faster when it is open,” he said. “You have a more vibrant ecosystem where everyone can contribute.”